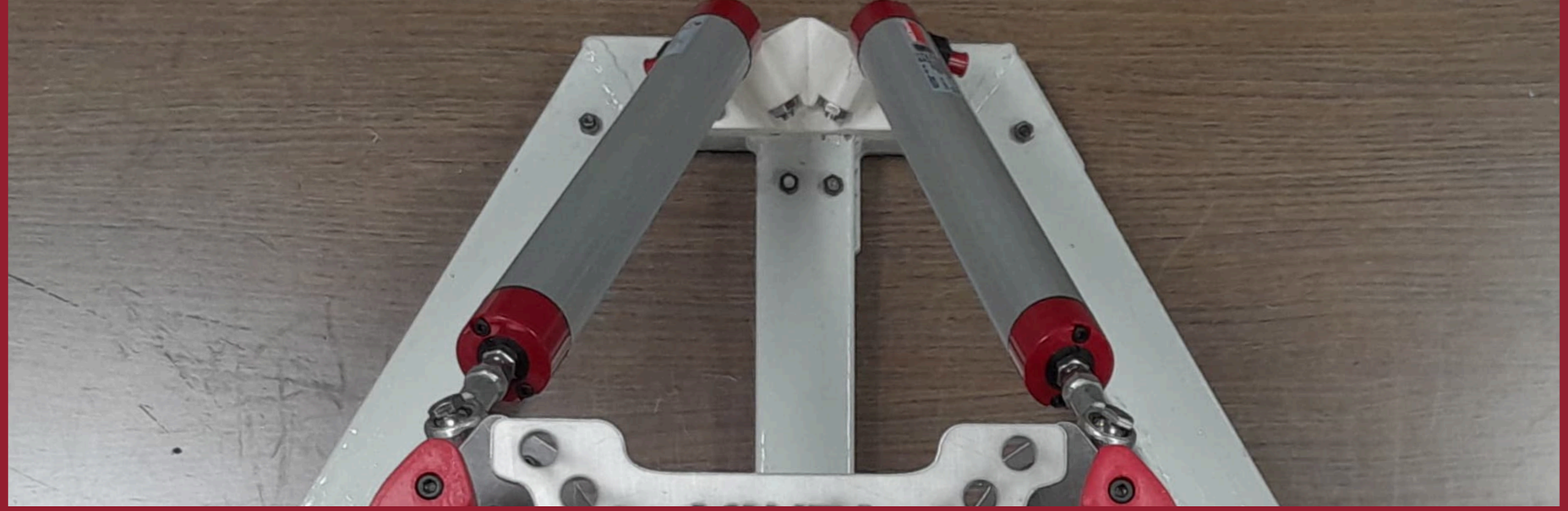
Design and development of 6-DoF Stewart Gough Platform for Potentiometer based Joystick

SHOBIT VERMA 21JE0893

Prof. Arun Dayal Udai
Assistant Professor
Department of Mechanical Engineering

B.TECH. FINAL YEAR FIRST PHASE PRESENTATION



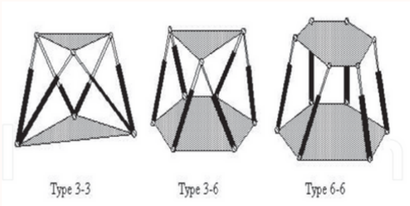
TASK - TO BUILD THIS ROBOT

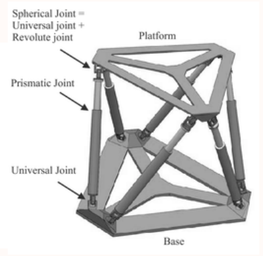


Introduction

- Stewart Gough Platform are the manipulators which having the end effector connected to the ground via several chains having actuations in parallel will have greater rigidity and superior positioning capability
- Used for applications where high load carrying capacity, good dynamic performance and precise positioning are of paramount importance, like flight simulator
- There are various configurations of this platform, but we would be looking at the 6 UPS (or 6 SPS) or commonly called the Stewart platform
- They are particularly proposed for good ratios of strength-to-weight.
- the topology also suffers from its complex and small work-space, kinematic singularities and inconsistent dexterity which also limits its flexibility

[1]

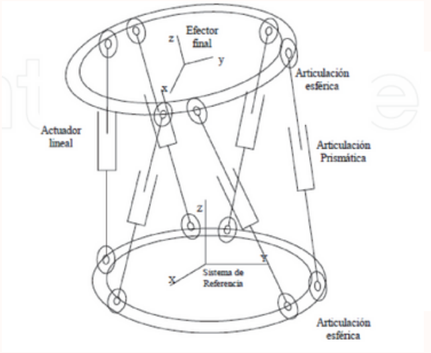




[2]

[3]

[4]



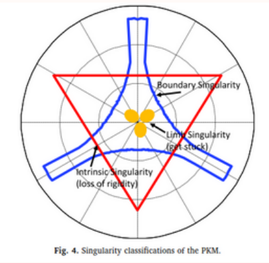


Fig. 4. Singularity classifications of the PKM.

REFERENCES

The Stewart platform manipulator: a review Bhaskar Dasguptaa,*, T.S. Mruthyunjaya [1]

Stewart-Gough Platform: Design and Construction with a Digital PID Controller Implementation Flabio Dario Mirelez-Delgado, José Ronaldo etc. [2]

Tlusty, J. Ziegert, S. Ridgeway, Fundamental comparison of the use of serial and parallel kinematics for machines tools, CIRP Ann. Manuf. Technol. 48 [3]

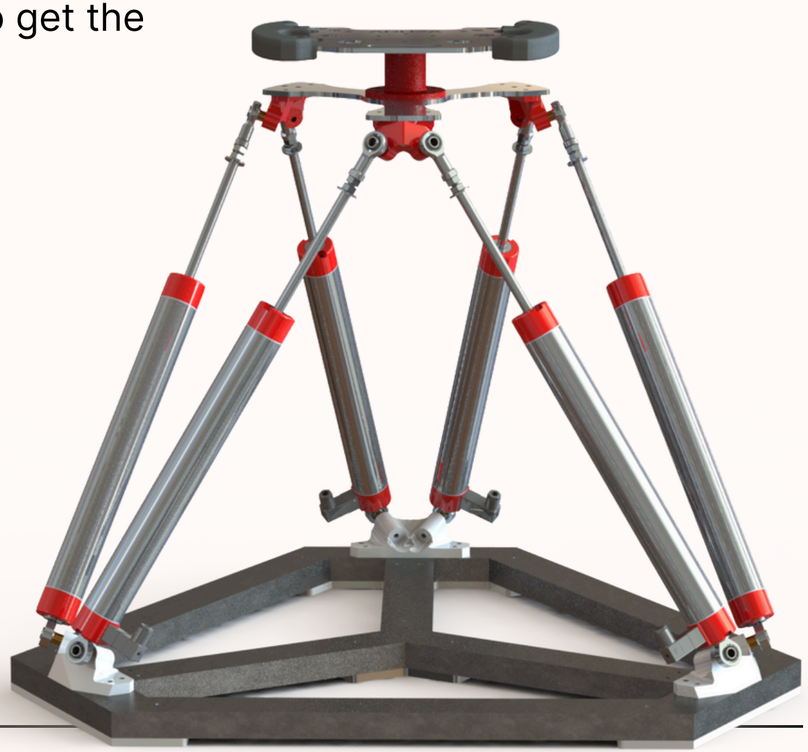
Optimization design for a compact redundant hybrid parallel kinematic machine Yen Ping-Lang a, Wang Ching-Henga, Lin Hao-Tingb, Hung Shuo-Sueic,d [4]

Objectives

- To build a 6 DOF Stewart Gough Platform with linear encoders, so that it can be used as a joystick for controlling larger platform
- As per this semester, main target was to finalize the CAD model, arrange all the materials and do manufacturing.
- Using Methods such as Waterjet, Milling, 3D printing, Metalworks etc to get the design into reality
- Reading literature and gaining knowledge about the platform
- Also create and execute elaborate plans to make build the Robot

Critical Challenges-

- Building the Kinematic skeleton of the bot
- Designing of the 3-D wedges
- Appropriate wedge angle for maximum workspace
- Acquiring materials and plan to manufacture it



Numerical formulation

Dimensions of our bot -

Base dia 777 mm, length 620 mm, Top Diameter 293mm, Inclination 60deg

The position vector of the ith joint on the top platform with respect to the ith actuator frame is given by

$$\mathbf{l}_i = \mathbf{t} + {}^{P}\mathbf{R}_O \cdot \mathbf{p}_i - \mathbf{b}_i$$

where ${}^{P}\mathbf{R}_O$ is the combined rotation matrix of the top frame with respect to the Stewart base frame O(X,Y,Z)

$${}^{P}\mathbf{R}_O = \mathbf{R}_z(\phi) \cdot \mathbf{R}_y(\theta) \cdot \mathbf{R}_x(\psi)$$

where,

$$\mathbf{R}_O = \begin{bmatrix} \cos\phi\cos\theta & -\sin\phi\cos\psi + \cos\phi\sin\theta\sin\psi & \sin\phi\sin\psi + \cos\phi\sin\theta\cos\psi \\ \sin\phi\cos\theta & \cos\phi\cos\psi + \sin\phi\sin\theta\sin\psi & -\cos\phi\sin\psi + \sin\phi\sin\theta\cos\psi \\ -\sin\theta & \cos\theta\sin\psi & \cos\theta\cos\psi \end{bmatrix}$$

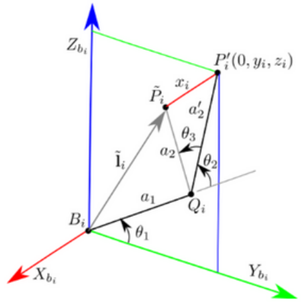
The vector Li was expressed in actuator frame Bi in order to obtain the joint angle of the ith actuator using

$$\tilde{\mathbf{l}}_i = [\mathbf{l}_i]_{B_i} = \mathbf{R}_z(\gamma_i) \cdot \mathbf{l}_i \qquad \tilde{\mathbf{l}}_i \equiv [x_i\ y_i\ z_i] \quad y_i = a_1\cos\theta_1 + a_2'\cos(\theta_1+\theta_2)$$
$$x_i = a_2\sin\theta_3 \qquad z_i = a_1\sin\theta_1 + a_2'\sin(\theta_1+\theta_2)$$

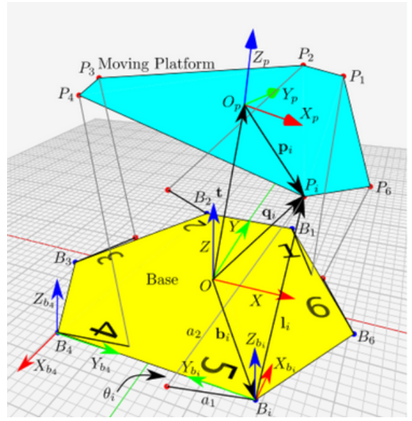
Therefore the position angles can be calculated as-

$$\theta_3 = \sin^{-1}\frac{x_i}{a_2} \text{ and } a_2' = a_2\cos\theta_3 \qquad \theta_2 = \cos^{-1}\left[\frac{y_i^2 + z_i^2 - (a_1^2 + a_2'^2)}{2a_1a_2'}\right]$$

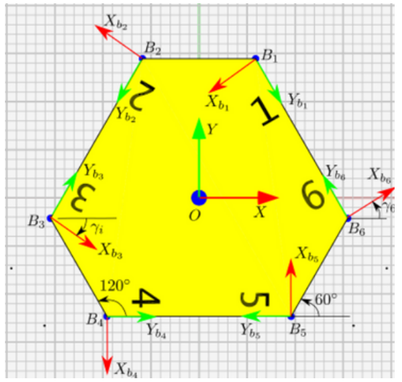
$$\theta_1 = \cos^{-1}\left(\frac{y_i}{A} - \varphi\right) \qquad \varphi = \tan^{-1}\left(\frac{a_2'\sin\theta_2}{a_1 + a_2'\cos\theta_2}\right)$$



Kinematic diagram of an isolated 3-R manipulator.

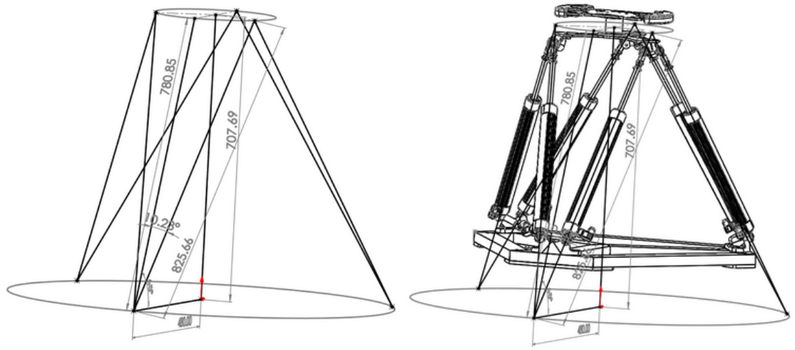


Kinematic diagram with symbols used for 6-RSS Stewart platform



Symbols	Description
θ_1	Angle of rotation for the first link about X_{bi} measured in ZY plane
θ_2	Angle subtended by the second link measured in ZY plane
θ_3	Angle subtended by the second link measured in XZ plane
$\tilde{P}_i$	$P_i \equiv (x_i, y_i, z_i)$ expressed in frame B_i
$\tilde{\mathbf{l}}_i$	$\mathbf{l}_i$ expressed in frame B_i
$P_i'(0, y_i, z_i)$	Projection of $\tilde{P}_i$ on ZY plane
a_2'	Projected length of a_2 on ZY plane

Additional symbols used

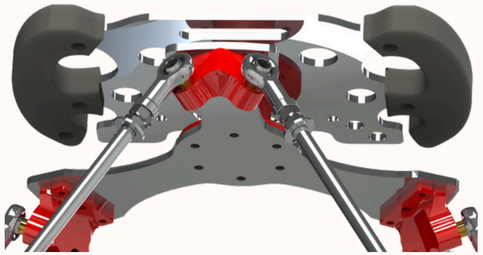


Kinematic Structure of our robot

REFERENCE

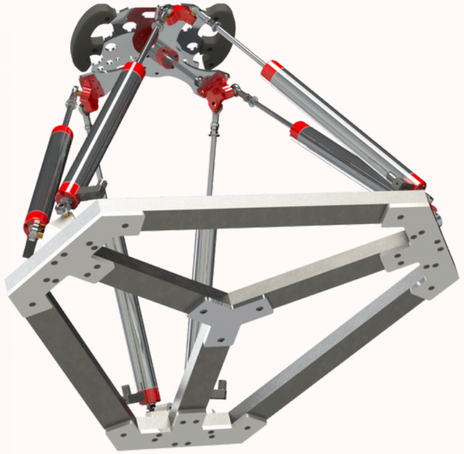
Kinematics and Statics of the Gough-Stewart Platform Jing-Shan Zhao * , Xiao-Cheng Sun and Song-Tao Wei

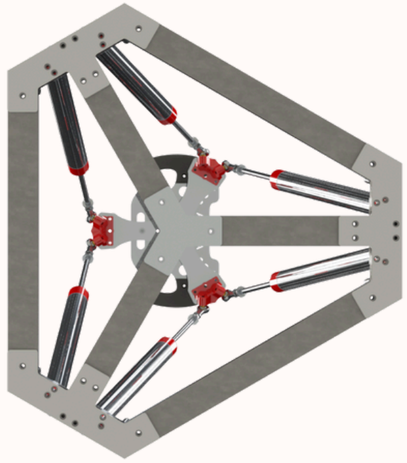


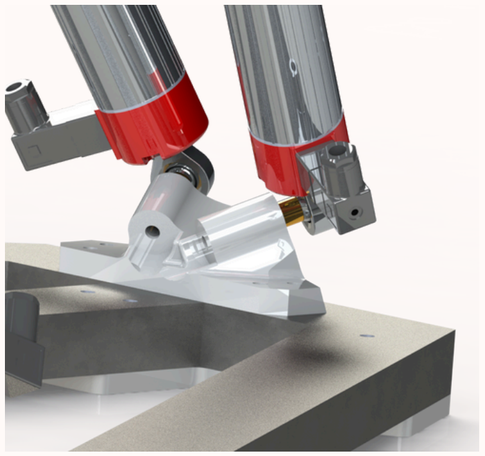






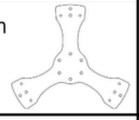
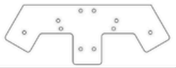
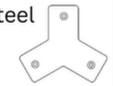
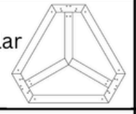


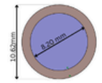




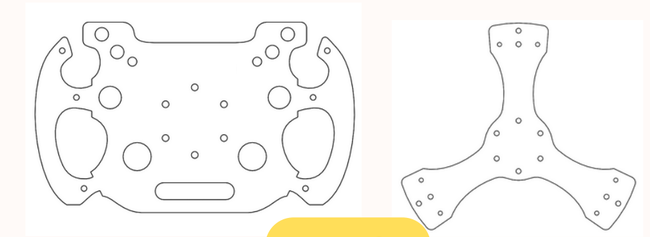
FINAL VERSION OF THE CAD DESIGN

List of components acquired

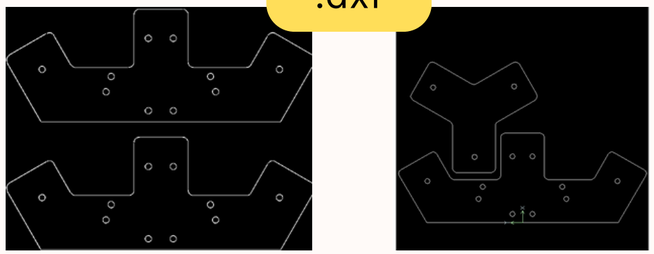
Part	Quantity	Material and dimenions	Remarks
Aluminum plate 	1	Aluminum plate 5mm thickness 300mm X 250mm	
Aluminum plate 	1	Aluminum plate 5mm thickness 300mm X 300mm	Solid aluminum, non scratch
Stainless Steel plate 	3	Stainless Steel plate 8mm thickness 320mm X 150mm	
Stainless Steel plate 	1	Stainless Steel plate 8mm thickness 160mm X 160mm	
Rectangular pipe 	4 or 5 m	Rectangular pipe 4m or 5m 2inch x 1 inch Cross Section	CS can be 2in x 2in
Red/White PLA 	1 roll or 1 roll each	PLA	Can try different colours
TPU/TPE 	1 roll (net quantity 250gms)	TPU/TPE	Black looks good?

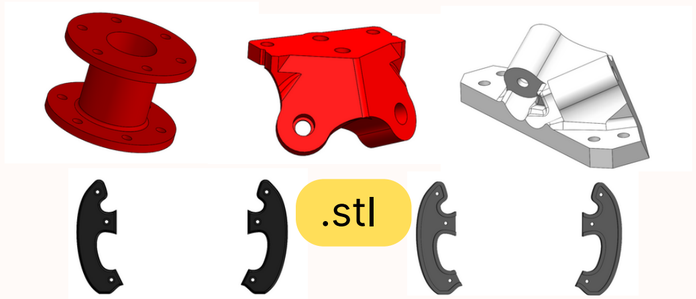
Bushings 	either pipe - 1m ***or*** 6+2 bushing 5mm long & 6+2 bushing 9mm long	Steel/Brass/bronze CS => 10.62 mm OD, 8.2 mm ID	(6+2 bushing means 6 required + 2 spare) Smallest ID = 8mm Largest OD = 11 or 12 or 13mm
M8 Shoulder Bolt 	6+2 pc with Shoulder length = 50mm 6+2 pc with Shoulder length = 30mm	Stainless Steel M8 X 50mm - 8pcs M8 X 30mm - 8pcs	(Length mentioned is Shoulder length)
M6 Bolts	17	M6 x 45mm	
	10	M6 x 40mm	
	17	M6 x 35mm	
	10	M6 x 20mm	
	26	M6 x 15mm	
M5 Bolts	17	M5 x 15mm	
M6 nuts	90		
M5 nuts	17		

Manufacturing of components



.dxf





.stl

Aluminum 5mm thickness

Waterjet



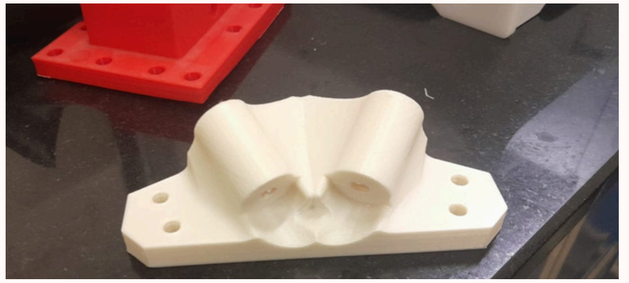
Nylon 8mm thickness

Waterjet + Counterbore Milling

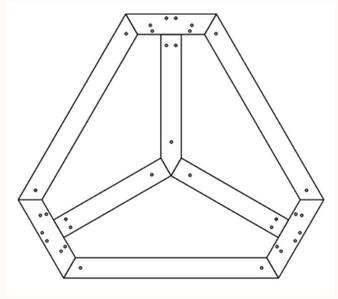


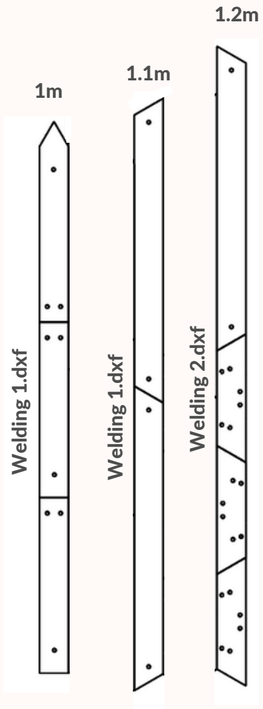
PLA / TPE

3D Printing



Manufacturing of base





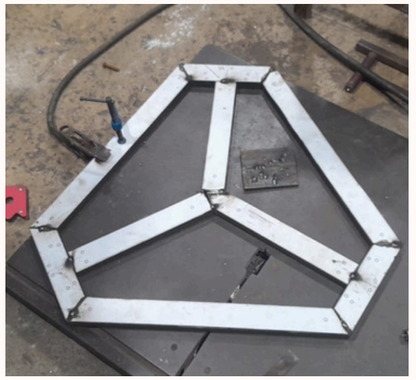
Design



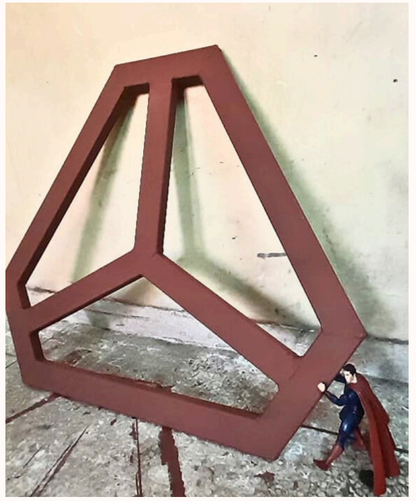


Flex pasting and cutting

Welding



Grinding

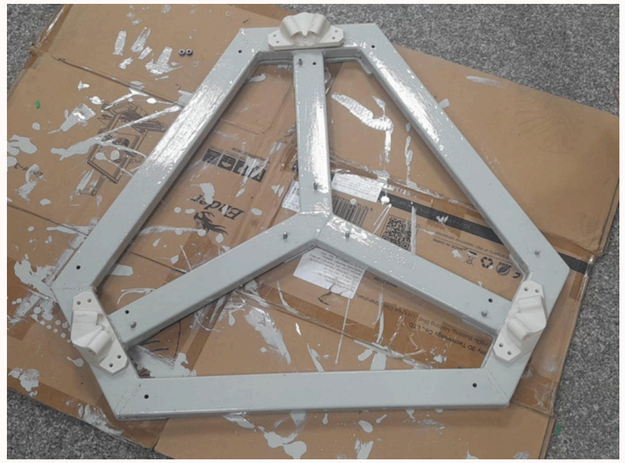


Sanding + red-oxide paint



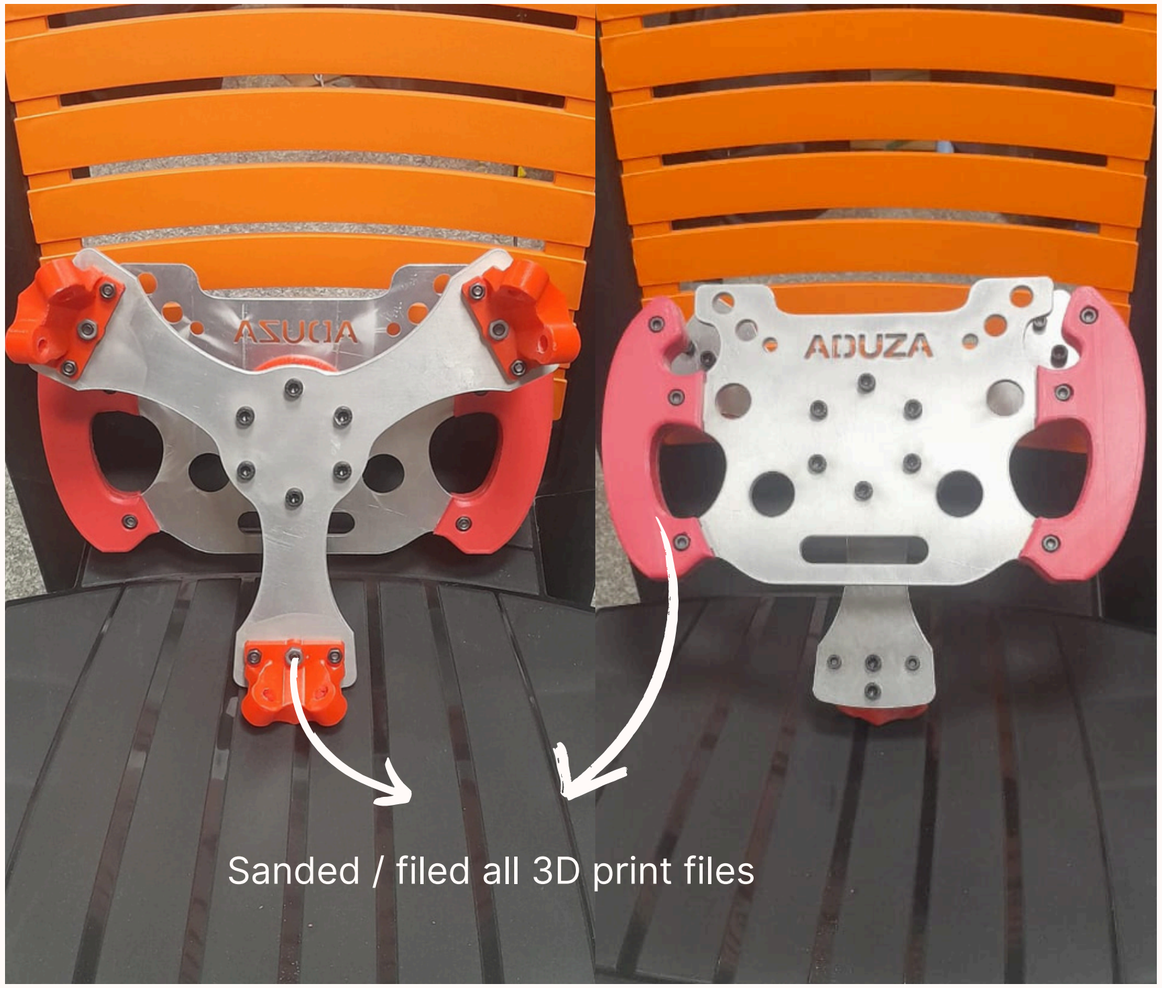
Markings for drilling

Assembly of parts





Enamel paint



Top assembly



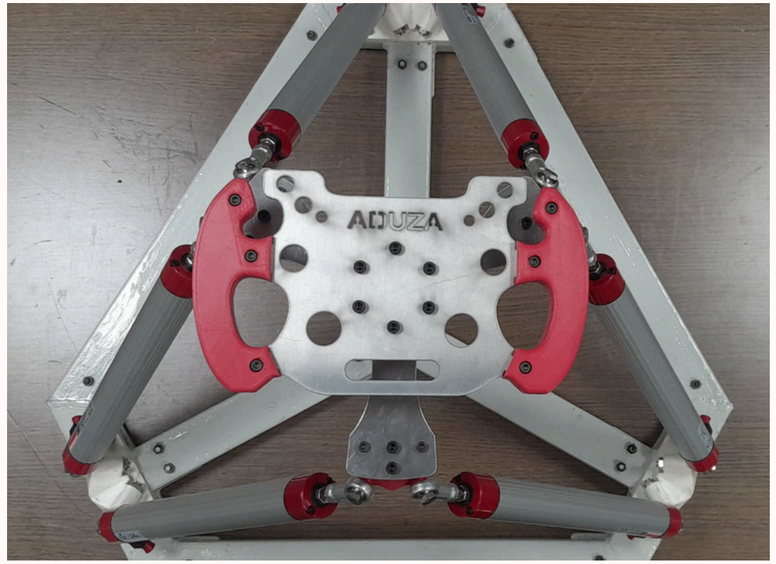
Finishing

Final Model





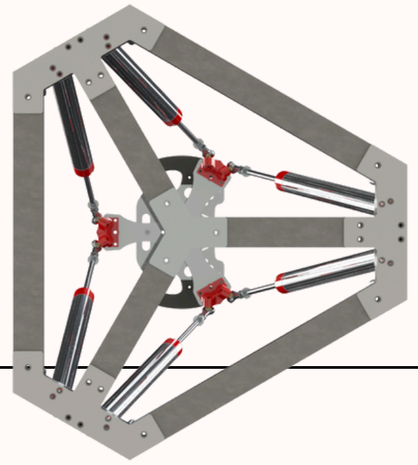
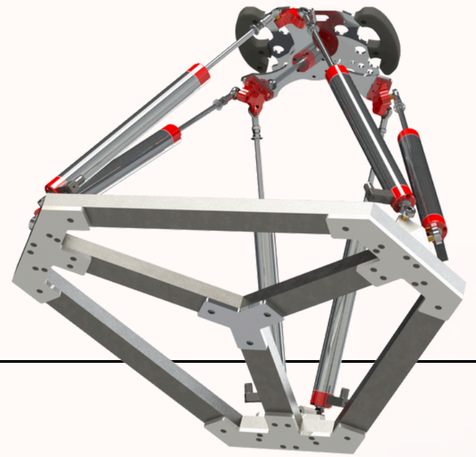
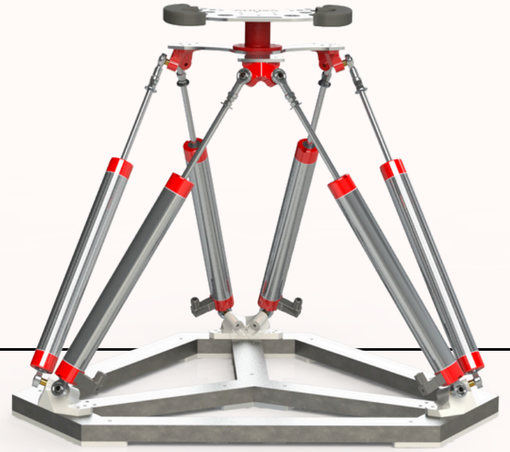
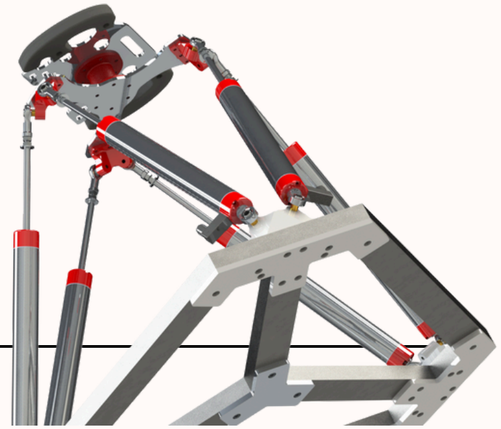






Results and Discussion

- Completed all the major milestones related to the building of the robot
- Developed the CAD model and plan to manufacture
- Learnt about the core aspects of the robot in general
- Successfully implemented Waterjet, Milling and 3D printing techniques
- Cutting, Grinding, filing, welding, etc were also used
- Learnt about various aspects of procuring materials and arrangement of components
- Assembled and finished the project

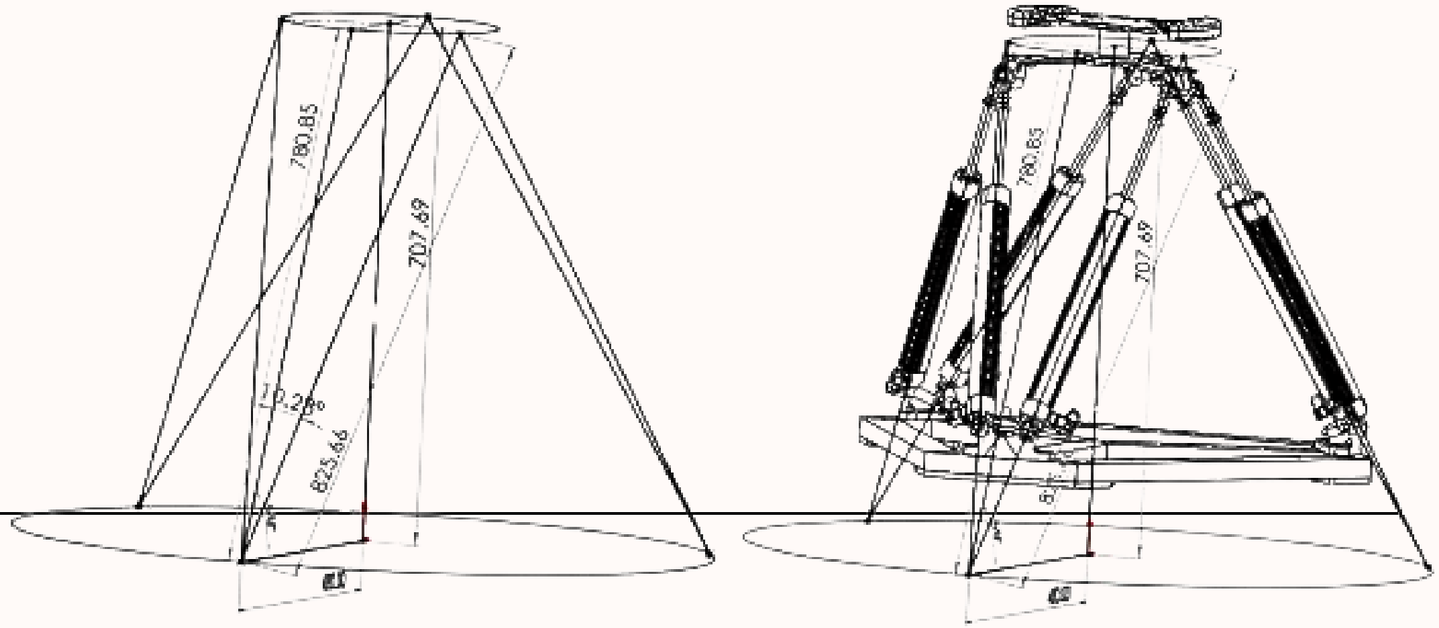





Conclusion

With the CAD model finished and the manufacturing process almost complete, It was a good learning experience. I would like to thank my Professor. Arun Dayal Udai Sir for his invaluable guidance that help me execute the tasks successfully. Moreover, I would also like acknowledge Ashish Siddharth and Nadeem for the crucial support throughout the stages of the project. This project helped me gain a lot of experience in building in the domains of building robots. I wish to further expand my knowledge and learn the control and circuit part of this project as well.

Future works

- Building the electronic circuit
- Learning controlling techniques
- Using ML to control the bot



THANK YOU !

